ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PRACTICE EXAM 4

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Paper 2 style (design and problem solving). Section A: objective questions (30) · Section B: a robot battery-monitor program (pseudocode, C code and test data), loops, arrays and input validation, writing functions, a Scratch game design and debugging (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Program code must be written in C unless stated otherwise. Assume that #include <stdio.h> is at the top of every program.
6. Write neatly and legibly. Indent your code so that it is easy to read.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The C operator that tests whether two values are equal is …

A = B == C != D =>

1.1.2 The printf format specifier for a float is …

A %d B %c C %f D %s

1.1.3 The loop best suited to repeating something a known number of times is the …

A for loop. B while loop. C do-while loop. D if statement.

1.1.4 In C, the operator && means …

A OR. B NOT. C AND. D equal to.

1.1.5 The correct statement to read an integer into the variable age is …

A scanf("%d", &age); B scanf("%d", age);
C printf("%d", &age); D read(age);

1.1.6 The array declared as int temps[7]; stores …

A one integer. B 6 integers. C 7 characters. D 7 integers.

1.1.7 A function that does not return a value has the return type …

A int B void C float D char

1.1.8 In Scratch, the block that repeats a set number of times is …

A forever. B repeat 10. C wait 1. D if … then.

1.1.9 A Raspberry Pi is a …

A small single-board computer. B printer.
C sensor. D motor.

1.1.10 A multi-line comment in C starts with …

A // B # C -- D /*

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 The statement i++; increases the value of i by 1.

1.2.2 A while loop always runs at least once.

1.2.3 Indenting code makes it easier to read.

1.2.4 The first element of an array in C has the index 1.

1.2.5 A function can be called many times in one program.

1.2.6 Testing a program with wrong input helps to find errors.

1.2.7 The statement printf("%d", 5.75); displays 5.75 correctly.

1.2.8 Scratch sprites can react when a key is pressed.

1.2.9 A C variable name may start with a number.

1.2.10 Pseudocode is written in plain language, not in a real programming language.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Repeats code while a condition is true	A	sprite
1.3.2	Chooses between two paths	B	function
1.3.3	Several values of the same type stored under one name	C	while loop
1.3.4	Named block of code that does one task	D	validation
1.3.5	Value given to a function when it is called	E	array
1.3.6	Checking that the input is sensible before using it	F	pseudocode
1.3.7	Finding and fixing errors in a program	G	if-else
1.3.8	Character in Scratch that runs scripts	H	debugging
1.3.9	Background picture of the Scratch stage	I	argument
1.3.10	Plain-language description of a program's steps	J	backdrop

(10)

TOTAL SECTION A: [30]

SECTION B

CASE STUDY. The college robot club needs small programs for its robots: a battery monitor, a lap timer and some helper functions, and a Scratch game to teach new members.

QUESTION 2: A BATTERY-MONITOR PROGRAM

2.1 Write pseudocode for a program that inputs the battery voltage of a robot and displays "Charge now" if it is below 6,5 V, "Low" if it is from 6,5 V up to below 7,0 V, and "OK" otherwise. (6)

2.2 Write the complete C program for Question 2.1. (8)

2.3 Draw up a table of FOUR test values for the program, including the boundary values, and the expected output for EACH. (4)

2.4 Explain why boundary values must be tested. (2)

[20]

QUESTION 3: LOOPS, ARRAYS AND VALIDATION

3.1 Write C code that declares an array for 5 lap times (decimal numbers) and uses a loop to read the 5 times from the keyboard. (5)

3.2 Write C code that calculates and displays the average lap time and the fastest (smallest) lap time. (6)

3.3 Write a loop that keeps asking for a lap time until a value greater than 0 is entered. (4)

3.4 Explain the difference between a for loop and a while loop, and say when you would use EACH. (3)

[18]

QUESTION 4: FUNCTIONS

4.1 Write a function cToF() that receives a temperature in °C as a float and returns it in °F (F = C × 9 ÷ 5 + 32). (4)

4.2 Write the statements in main() that call cToF() for 25 °C and display the answer. (3)

4.3 Write a void function printLine() that receives an integer n and prints a line of n stars (*). (5)

4.4 Give TWO benefits of using functions. (2)

4.5 Explain the difference between a parameter and an argument. (2)

[16]

QUESTION 5: SCRATCH AND DEBUGGING

5.1 Describe the scripts for a Scratch game in which a robot sprite moves with the arrow keys and says "Battery low!" for 2 seconds when it touches a red area. Name the blocks you would use. (6)

5.2 The program below should count to 5 and display the count, but it contains errors. Find FOUR errors and write the correction for EACH. (8)

```
int Main() {
  int count = 0
  for (i = 0; i < 5; i++) {
    count = count + 1;
  printf("%f", count);
```

```
  return 0;
}
```

5.3 Write down the output of the program once it has been corrected (with the printf after the loop). (2)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER